CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 1: Orienting Yourself: The Use of Coordinates

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Name the quadrant in which each point lies: $(-3, 5)$, $(4, -1)$, $(-2, -7)$.
2. Write the coordinates of the point on the y-axis that is 6 units below the origin.
3. Find the distance between $(-2, 5)$ and $(7, 5)$.
4. Find the distance between $(2, -1)$ and $(-4, 7)$.
5. Find the perimeter of the triangle with vertices $(0, 0)$, $(8, 0)$ and $(8, 6)$.
6. Check whether $(-2, -3)$, $(1, 1)$ and $(7, 9)$ are collinear.
7. Write the images of $(4, -7)$ after reflection in the x-axis and in the y-axis.
8. Find the mid-point of the segment joining $(-6, 3)$ and $(2, -9)$.
9. $M(1, 2)$ is the mid-point of $A(-3, 5)$ and B. Find B.
10. A circle has its centre at the origin and radius 13 units. Show that $(5, 12)$ lies on it, and say where $(9, 9)$ lies.

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Answer Key

1. II, IV, III
2. $(0, -6)$
3. $|7 - (-2)| = 9$ units
4. $\sqrt{36 + 64} = 10$ units
5. $8 + 6 + 10 = 24$ units
6. The distances are 5, 10 and 15, and $5 + 10 = 15$, so they are collinear.
7. $(4, 7)$ and $(-4, -7)$
8. $(-2, -3)$
9. $(5, -1)$
10. $\sqrt{25 + 144} = 13$, so $(5, 12)$ is on the circle. $\sqrt{162} \approx 12.7 < 13$, so $(9, 9)$ is inside.

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